

Enda Eyesus Community Digital Ecosystem Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the technical specifications, architectural constraints, and functional requirements for the Minimum Viable Product (MVP) of the Enda Eyesus Community Digital Ecosystem and Learning Management System (LMS).

This document serves as the primary technical blueprint for the full-stack development team. It will also be used by the Enda Eyesus Gbi Gubae Executive Committee (ሥራ አስፈጻሚ ጉባኤ) as a validation tool to ensure the software aligns with the spiritual, administrative, and educational workflows of the campus ministry.

1.2 Document Conventions

To ensure clarity across both technical and non-technical stakeholders, this document adheres to the following conventions:

- **Bold Text:** Used to indicate specific database tables, schema models, or critical architectural components (e.g., **User**, **ServiceClass**).
- *Italics:* Used for native Amharic/Ge'ez terminologies and official departmental titles as defined in the Gbi Gubae bylaws.
- **Terminologies:**
 - *Gbi Gubae:* The university campus ministry.
 - *Gubae Abew:* The foundational course required for service class membership.
 - *Gubae Hawaryat / Gubae Ecclesiae:* Advanced phase-based educational tracks.
 - *RBAC:* Role-Based Access Control, the security paradigm governing what actions a user can perform.

1.3 Project Scope

The Enda Eyesus platform is a centralized web application designed to digitize the administrative, educational, and communication workflows of the campus ministry. The system operates on a multi-tiered access architecture:

- **Public Access (No Login Required):** The system serves as a public-facing portal for general information. Unauthenticated users can access the "About the Gbi Gubae" page, read Public Announcements, and browse the Public Library.
- **Automated Membership & Registration:** Membership into official Service Classes is strictly regulated. The system will feature an automated verification engine. When a user applies for membership, the system cross-checks their identity against a pre-seeded database of *Gubae Abew* graduates managed by the Member Affairs department. If a match is found, the registration is automatically processed; if not, it is flagged for manual review or rejection.
- **Phase-Based LMS (Course Registration):** Access to the Learning Management System is distinct from service class membership. Users must register for specific courses (*Gubae Abew*, *Gubae Hawaryat*, *Gubae Ecclesiae*) sequentially, governed by a phase-based prerequisite system. **Important:** The Phase 1 (*Gubae Abew*) course pages are fully open to any user authenticated with a Tier 1 (Public Access Account). No membership verification is required to view or engage with *Gubae Abew* content. However, completion of *Gubae Abew* is tracked and is a prerequisite for accessing Phase 2 (*Gubae Hawaryat*) and Phase 3 (*Gubae Ecclesiae*), which do require full membership.
- **Strict Role-Based Access Control (RBAC):** Privileges are granularly restricted. While an ordinary member can read, react, and comment on class-specific announcements or download library files, only designated Roles (e.g., Secretariat, Class Admins, System Admins) possess the CRUD (Create, Read, Update, Delete) privileges required to post updates, upload resources, or access administrative dashboards.

Out-of-Scope (What the system is NOT):

- **Public Social Network:** The system will *not* feature open, user-generated public posting outside of official LMS course discussions. Announcements are strictly one-way or semi-restricted communications from leadership.
- **Heavy Media Hosting:** To optimize database performance, the system will *not* host heavy video files directly. It relies strictly on external integrations (e.g., YouTube IFrame API).
- **Visual Course Builder:** The system will **not** include any automated visual course builder. All course content creation follows a manual submission and implementation workflow (see Section 3.3.2).

1.4 References

The design and business logic of this system are derived from the following official Enda Eyesus documents and technical frameworks:

1. የእንዳ ኢየሱስ ግቢ ጉባኤ የውስጥ ደንብና የአሠራር መመሪያ (Internal Bylaws and Operational Guidelines).
2. የእንዳ ኢየሱስ ግቢ ጉባኤ የአገልግሎት መዋቅርና የስራ መመሪያ (Service Structure and Functional Directive).
3. Enda Eyesus Product Requirements Document (PRD) V1.0.

2. Overall Description

2.1 Product Perspective

The Enda Eyesus Community Digital Ecosystem is a greenfield project designed to replace a fragmented legacy ecosystem of manual paper-based registrations, physical exams, and scattered social media communication. It is an independent, centralized web application, but it acts as a composite system by heavily relying on external cloud integrations to maintain performance.

Rather than building an isolated monolith, the system interacts with:

- **Google Drive API:** Acts as the primary file storage and document management solution for the Resource Library and LMS reading materials. The Enda Eyesus database only stores the shareable links.
- **YouTube IFrame Player API:** Serves as the exclusive video hosting and streaming engine for LMS lectures and spiritual content, completely offloading bandwidth requirements from the application servers.

(See Section 6 for detailed interface requirements.)

2.2 User Classes and Characteristics

The system serves a diverse demographic of university students, characterized by high mobile phone usage but varying levels of technical literacy. The platform enforces strict Role-Based Access Control (RBAC) across the following user classes.

Role	Brief Description
Global Admin (System Administrator)	Full CRUD access across the entire system.
Secretariat (Chairperson, Vice, Secretary)	Global view of all departments, exclusive rights to post Public Announcements, view system-wide reports.
Service Managers (e.g., Member Affairs, Education)	Department-scoped control; manage sub-classes, assign members within their department.
Teachers	Submit course content to Education Class for verification; after approval, content is manually implemented by technical team; also publish quizzes, grade manual exams, view student progress.
Members	Full access to authenticated areas: enroll in LMS (Phases 2 & 3), download resources, react/comment on department announcements.
Candidates / Pre-Members	New sign-ups awaiting verification; have full access to Gubae Abew (Phase 1) course pages but restricted from advanced LMS tracks and full membership features.

2.3 Operating Environment

To ensure maximum accessibility for university students, the system will operate under the following environmental parameters:

- **Client-Side (User Interface):** A responsive, mobile-first web application accessible via modern web browsers (Chrome, Safari, Firefox, Edge). There is no native iOS/Android app requirement for Version 1.0; the web app must function flawlessly on mobile viewport sizes.
- **Server-Side:** Hosted on managed cloud infrastructure (e.g., AWS, Azure, GCP, or platform-as-a-service) utilizing Node.js/NestJS. Database: A cloud-hosted PostgreSQL relational database.
- **Centralized Diagnostics & Performance Monitoring:** The system must integrate error tracking and application performance monitoring (APM) with automated alerting for unhandled exceptions. Personal data must be masked in logs.
- **Database:** A cloud-hosted PostgreSQL relational database.

2.4 Design and Implementation Constraints

The architecture is strictly bound by the following technical choices to guarantee zero-cost scalability and maintainability:

- **Zero-Blob Database Storage:** The PostgreSQL database is strictly forbidden from storing heavy media files (images, PDFs, videos). Profile pictures and lightweight UI images must be heavily compressed client-side before upload, while all other media must use Google Drive or YouTube links.
- **Relational Depth Limits:** To prevent N+1 query performance issues, the organizational hierarchy is constrained to a maximum of two tiers: **Department -> Sub-Class**. Infinite recursive sub-classing is disallowed.
- **LMS Content Standard:** The system does **not** include any automated Visual Course Builder. Instead, all course content follows a manual submission and implementation workflow: Teachers submit content → Education Class verifies → Technical Team manually implements as web pages and embeds them in the designated course delivery location.
- **Immutability of Ledger:** The announcement feature is restricted to a top-down official communication channel, eliminating the need for automated community strike systems.

2.5 Assumptions and Dependencies

The successful deployment and operation of this project assume the following:

- **Internet Access:** The target audience (university students) has reliable access to the internet and personal smart devices.
- **External API Stability:** The YouTube IFrame API and Google Drive API will remain stable, free, and accessible without severe rate limits for the projected volume of users.
- **Initial Data Integrity:** The Member Affairs department possesses an accurate, digitized list of past *Gubae Abew* graduates that can be used to successfully seed the database for the automated membership verification engine.

2.6 Use Cases (Key Scenarios)

UC ID	Use Case	Primary Actor	Brief Description
UC-01	Student Self-Registration	Candidate	Tier 1 sign-up → Tier 2 university data → automated verification → member
UC-02	Manual Verification Override	Member Affairs Manager	Review PENDING_REVIEW profile, validate manually, approve/reject
UC-03	Course Enrollment (Phase 2 or 3)	Member	Browse active batches, register for next available course meeting prerequisites (Gubae Abew completion required for Phase 2)
UC-04	Submit Course Content for Implementation	Teacher	Prepare content → submit to Education Class for verification → Education Manager approves → Technical Team implements as web page → course delivered
UC-05	Batch Graduation	Education Manager	Close batch → calculate final grades → generate certificates → offboard students
UC-06	Post Internal Announcement	Service Manager	Compose announcement → system restricts visibility to department members
UC-07	Assign Pastoral Roles	Member Affairs Manager	Assign Repentance Father/Deacon/Spiritual mentors to a member profile
UC-08	Access Gubae Abew Course	Any Tier 1 User	Login with Tier 1 account → navigate to Gubae Abew course page → full read access to all content (no membership required)

3. System Features

3.0 Requirement Prioritization

Each functional requirement is labeled with a priority according to the following scheme:

- **P0 (Critical):** Must be implemented for MVP launch.
- **P1 (High):** Should be implemented for MVP; can be post-MVP if necessary.
- **P2 (Medium):** Planned for post-MVP release 1.
- **P3 (Low):** Future enhancement.

Default priority for all requirements is P1 unless otherwise noted.

3.1 System Feature: Registration & Onboarding Workflow (Prefix: REG)

3.1.1 Description

This module governs a unified, two-tiered authentication and onboarding pipeline designed to manage user identities cleanly without database pollution.

- **Tier 1 (Public Access Account):** Establishes a permanent digital identity using a **Full Name, Email Address, and Password** credential set. This allows casual visitors or unverified students to secure an account, access public-facing materials, **and fully view/engage with the Gubae Abew (Phase 1) course pages** without requiring full fellowship membership activation.
- **Tier 2 (Membership Activation):** Reuses the active authenticated Tier 1 session. The system automatically inherits the user's core identity fields. The user only needs to provide supplementary university data to run the automated validation process against the graduate ledger.

3.1.2 Functional Requirements

FR-REG-01 [P0]: The system must provide an initial registration form requiring exactly three inputs: Full Name, Email Address, and a secure Password.

FR-REG-02 [P0]: Upon form submission, the system must hash the password, instantiate a record in the **users** table, assign a baseline role of USER, and issue a secure session token.

FR-REG-03 [P0]: The system must grant users authenticated with a basic USER token immediate read-write-react access to the Public Announcements module, the general "About" directory, the public index of the Resource Library, and **unrestricted access to all Gubae Abew (Phase 1) course pages**.

FR-REG-04 [P1]: The system must log session activity and unique login timestamps bound to these credentials to provide the executive committee with analytics on public engagement and reach.

FR-REG-05 [P0]: When an authenticated user initiates the full fellowship membership application, the system must inherit the `id`, `full_name`, and `email` directly from the active session context. The frontend must display these fields as immutable, read-only values.

FR-REG-06 [P0]: The fellowship registration portal must require: Academic Department, Current Academic Year (Tiers 1–5), assigned Dorm Block, and Dorm Room number.

FR-REG-07 [P1]: The registration form must provide a dropdown selector for the user's "Preferred Service Class". A visible informational link must route to a detailed profile summary page explaining each department.

FR-REG-08 [P0]: Upon submission, the backend must execute an automated query checking the provided Full Name against the pre-seeded ledger of *Gubae Abew* graduates.

FR-REG-09 [P0] (Match Found): If verification succeeds:

- Present a digital "Orientation Checklist".
- Upon acknowledgment, atomically update the user's role to MEMBER.
- If the user leaves the dropdown blank, automatically assign the pre-seeded 'Default Class' UUID to `service_class_id` (NULL is prohibited for verified members).
- Trigger an immediate notification on the Member Affairs Manager's dashboard.

FR-REG-10 [P0] (No Match Found): If verification fails:

- Place the application into `PENDING_REVIEW` state; keep Tier 1 public privileges active (including *Gubae Abew* access).
- Route the complete profile to the Member Affairs review queue for manual handling.

3.2 System Feature: Official Announcements System (Prefix: ANN)

3.2.1 Description

A controlled, top-down communication ledger designed to replace scattered social media updates. This feature prioritizes official organizational updates over peer-to-peer social networking.

3.2.2 Functional Requirements

FR-ANN-01 [P0]: Creation, modification, and deletion of Public Announcements is restricted to **Secretariat** roles.

FR-ANN-02 [P0]: Service Managers can compose internal announcements; backend enforces `targetClassId` so only matching department members see them.

FR-ANN-03 [P1]: Supports headers, bold, blockquotes, lists, and emoji rendering.

FR-ANN-04 [P1]: Both USER and MEMBER roles can react (Like indicates approval; Star indicates bookmark/save) and comment (1-level threading) on visible announcements.

FR-ANN-05 [P0]: Original author and `SECRETARIAT_CHAIRMAN` / Global Admins can edit or hard-delete any announcement, comment, or reaction.

3.3 System Feature: LMS Engine (Prefix: LMS)

3.3.1 Description

Manages phase-based theological education (Gubae Abew → Gubae Hawaryat → Gubae Ecclesiae) with strict prerequisites and a cyclical Batch ($\mathbb{H}\mathbb{C}$) isolation model.

Important distinction:

- **Gubae Abew (Phase 1):** Fully open to any Tier 1 (Public Access Account) user. No membership verification required. Users can view all content, track progress, and complete quizzes. Completion is recorded and serves as prerequisite for Phase 2.
- **Gubae Hawaryat (Phase 2) & Gubae Ecclesiae (Phase 3):** Require formal Tier 2 Membership status. Only verified members can enroll and access these advanced tracks.

Furthermore, the system operates on a cyclical Batch ($\mathbb{H}\mathbb{C}$) isolation model, ensuring that only students belonging to the active, ungraduated academic batch can access course contents and assessments for each phase. Upon graduation of a batch, the system seals their records, issues digital certificates, and offboards their active states.

3.3.2 Functional Requirements

FR-LMS-01 [P0]: Strict linear prerequisites:

- Gubae Abew completion is the absolute requirement for entering Gubae Hawaryat.
- Gubae Hawaryat completion is the absolute requirement for entering Gubae Ecclesiae.

- If a student attempts to bypass this sequence, the UI must block registration and the backend must reject the API request with an HTTP 403 Forbidden status.

FR-LMS-02 [P0]: Batch (፡፡) Lifecycle Control: Education Manager can create, activate, or terminate a Batch. Only one batch per course track can be ACTIVE at any given time.

FR-LMS-03 [P0]: Active Batch Visibility Guard: Only students linked to the current ACTIVE batch for a given course track can access lessons, quizzes, and submission portals for that track.

FR-LMS-04 [P0]: Graduation and Final Grade Submission: "Close Batch & Graduate" utility calculates final cumulative grades and converts student status from ACTIVE_STUDENT to GRADUATED.

FR-LMS-05 [P0]: Automated Certificate Generation & Offboarding:

- Upon successful batch graduation, generate a verifiable digital completion record for each passing student. The certificate must include a unique hash that can be verified on a public certificate page accessible via QR code.
- Revoke interactive access to that specific course track for graduated students.

FR-LMS-06 [P0] – Manual Course Submission Workflow :

The system must **not** provide any automated visual course builder. Instead, the following manual workflow shall be implemented:

1. **Teacher Submission:** Teachers prepare course content (text, images, video links, document links, quiz questions) using any external tool (e.g., word processor) and submit the material through a dedicated submission interface in the system. The system stores the submission as a "Draft" content package linked to the specific course track and batch.
2. **Education Class Verification:** The Education Manager receives a notification of pending content. The manager reviews the submitted material for theological accuracy, completeness, and compliance with fellowship guidelines. The manager can:
 - Approve the content → status changes to "Approved" and notifies the Technical Team.
 - Reject with feedback → status returns to "Draft" for teacher revision.
3. **Technical Team Manual Implementation:** Upon approval, the Technical Team (who have a dedicated system role with appropriate access) retrieves the approved content package. The team manually creates structured web pages (HTML/CSS/JS) that present the course content in a clean, responsive, and branded layout. These pages are then uploaded to the system's designated course delivery location (e.g., a specific URL path or content management system).
4. **Course Publication:** Once the Technical Team marks the implementation as complete, the system sets the course lesson status to "Published" and makes it visible to students enrolled in the active batch of that course track. The system stores only the URL or reference to the manually implemented web page in the database (e.g., in the completed_lessons JSONB field or a dedicated lesson_page_url column).

FR-LMS-07 [P0]: The content pipeline must support status tracking: Draft (teacher working) → Submitted (pending Education Class review) → Under Review (Education Class evaluating) → Approved (ready for technical implementation) → Implementation in Progress (Technical Team working) → Published (live to students). Rejected items return to Draft with feedback.

FR-LMS-08 [P1]: Dual assessment engine: auto-graded Multiple Choice Questions (MCQ) quizzes (configured by teachers during submission) and manually-graded open-ended paragraph exams handled via a Teacher Grading Dashboard.

FR-LMS-09 [P0] – Gubae Abew Open Access: The system must grant any authenticated USER (Tier 1) full, unrestricted read access to all Gubae Abew (Phase 1) course pages, including lessons, videos, documents, and quizzes. No membership or batch enrollment is required for viewing, although quiz completion and progress tracking are recorded per user. Enrollment in an active Gubae Abew batch is automatically granted upon first access.

3.4 System Feature: Resource Library (Prefix: LIB)

3.4.1 Description

Centralized, searchable digital repository using external cloud storage links (zero-blob database).

3.4.2 Functional Requirements

FR-LIB-01 [P0]: Direct upload of binary files (PDF, PPT, etc.) into the database is rejected.

FR-LIB-02 [P0]: Only public Google Drive URLs are accepted; assets are rendered inside iframes or a PDF viewer component.

FR-LIB-03 [P1]: A weekly background job performs HEAD requests on all Google Drive URLs; broken links are marked LINK_BROKEN and the system must send an in-app notification to the Library Administrator.

FR-LIB-04 [P0]: Categorization: Spiritual (መንፈሳዊ), Academic (አካዳሚክ), Others (ሌሎች).

FR-LIB-05 [P1]: Recursive filtering by Department → Academic Year → Course ID → Document Type (Textbooks / Past Exam Papers).

FR-LIB-06 [P1]: Authenticated users can like, download, or react to any library item.

FR-LIB-07 [P2]: Real-time aggregate counts of likes/interactions displayed on each file card.

3.5 System Feature: Member Management & Department Assignment (Prefix: MEM)

3.5.1 Description

Governs final deployment, role modification, and spiritual care assignment of verified members. Member Affairs has global authority; Service Managers have scoped authority.

3.5.2 Functional Requirements

FR-MEM-01 [P0]: Notification to Member Affairs when a user becomes a full member.

FR-MEM-02 [P0]: Member Affairs can approve preferred class, override assignment, or move user out of Default Class.

FR-MEM-03 [P0]: Member Affairs can alter identity fields, transition users between service classes, and update spiritual profiling.

FR-MEM-04 [P1]: Notification to the receiving department's Service Manager after class placement.

FR-MEM-05 [P0]: Service Managers cannot view/modify members of other departments.

FR-MEM-06 [P1]: Service Managers can assign sub-class roles (Sub-Chair, Sub-Vice, Sub-Secretary); cannot modify core member fields.

FR-MEM-07 [P0]: Education Manager can grant/revoke TEACHER role.

FR-MEM-08 [P0]: Member Affairs can assign Repentance Fathers, Coordinator Deacons, Spiritual Fathers/Mothers.

FR-MEM-09 [P0]: Every student profile displays their assigned spiritual mentors with name, contact, and profile picture.

FR-MEM-10 [P0]: As soon as a service class and sub-role are assigned, the user's access matrix is recalculated and they gain all associated privileges.

3.6 System Feature: Centralized Notification Engine (Prefix: NOT)

3.6.1 Description

Cross-cutting signaling core for real-time alerts and transaction warnings.

3.6.2 Functional Requirements

FR-NOT-01 [P1]: In-app notifications plus asynchronous email (SMTP) for urgent workflows.

FR-NOT-02 [P1]: Frontend displays unread counts without page reload.

FR-NOT-03 [P1]: Database tracks read/unread per notification; supports MARK_AS_READ, ARCHIVE_ALL, CLEAR_DISMISS.

FR-NOT-04 [P2]: Notifications contain a targetRoute; clicking navigates directly to the relevant workspace.

3.7 System Feature: Role-Based Access Control (Prefix: RBAC)

3.7.1 Description

Centralized access control interceptor validating request signatures against a permissions hierarchy.

3.7.2 Functional Requirements

FR-RBAC-01 [P0]: The system must enforce the following permissions matrix (single source of truth):

Role Identifier	Permitted System Actions & Structural Boundary Limits
SECRETARIAT_CHAIRMAN	Super administrative access; can override any record, author public announcements, assign/revoke all secretariat and service manager roles; can securely transfer own root role.
SECRETARIAT_VICE	Global write/read access; manage Public Announcements, full CRUD on Resource Library; read-only view of member data.
SECRETARIAT_SECRETARY	Same as Vice but also read-only on Member Profiles.
SERVICE_MANAGER	Scoped to own department: author internal announcements, manage sub-class roles, suspend local members. Education

	Manager can grant TEACHER; Member Affairs Manager can override membership validation and assign pastoral roles.
TEACHER	Submit course content to Education Class, view student progress, grade manual exams. No direct content publishing or visual builder.
MEMBER	Read/react/comment on public & departmental announcements; enroll in active LMS tracks (Phases 2 & 3); update own profile; download library resources.
USER	Public visitor: view About page, Public Announcements, Resource Library. Full unrestricted access to Gubae Abew (Phase 1) course pages. No access to Phases 2/3 or department-specific features.

FR-RBAC-02 [P0]: Root URL (/) serves a public landing page with fellowship info and authentication modal. Unauthenticated visitors can browse public content (About page, Public Announcements, Public Library). Authentication (Tier 1) is required to access any member-only features or the Gubae Abew course pages.

FR-RBAC-03 [P0]: Every request validates HttpOnly cookie token; unauthorized access returns HTTP 403 and logs violation.

FR-RBAC-04 [P0]: On first schema build, a seeding script runs. If environment variables INITIAL_CHAIRMAN_EMAIL and INITIAL_CHAIRMAN_PASSWORD are present, create a SECRETARIAT_CHAIRMAN user, bypassing *Gubae Abew* validation.

FR-RBAC-05 [P0]: The chairman can then assign other executive roles.

3.8 Business Rules Catalog

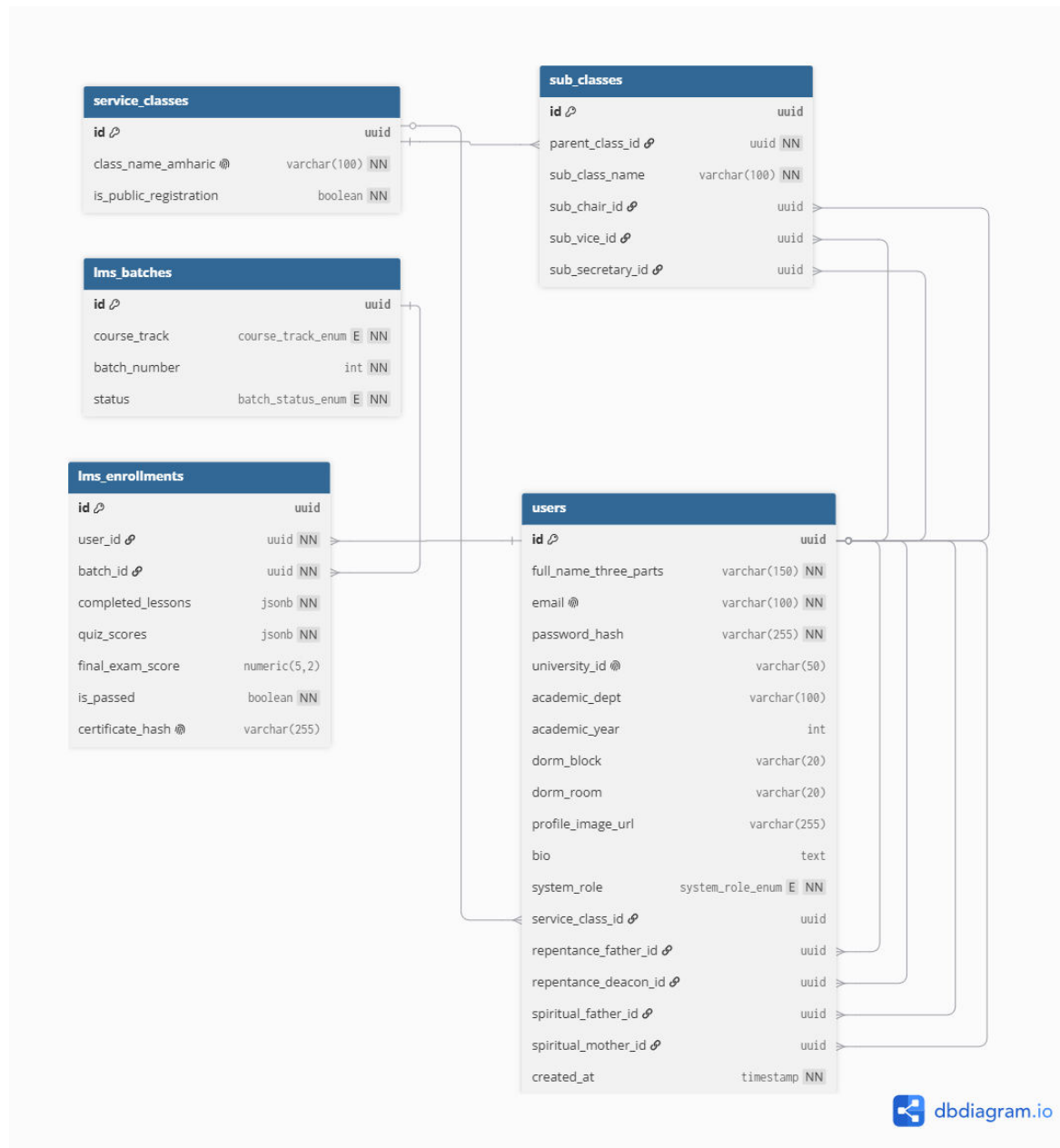
ID	Rule	Enforcement Point
BR-01	Only one batch per course track can have status ACTIVE simultaneously.	LMS batch creation/activation
BR-02	Maximum two organizational tiers: Department → Sub-Class. No further nesting.	Database schema and UI
BR-03	A verified member's service_class_id must never be NULL (default class assigned if none selected).	Registration verification
BR-04	Gubae Ecclesiae registration requires is_passed = TRUE for Gubae Hawaryat.	Course enrollment API

BR-05	Public announcements can only be created by SECRETARIAT roles.	RBAC guard
BR-06	Gubae Abew course pages are accessible to any authenticated USER (Tier 1) without membership.	Route guard / RBAC
BR-07	No automated visual course builder; all course content follows manual submission → verification → implementation workflow.	System design constraint

4. Data Requirements

4.1 Logical Data Model

The database design follows a strict relational structure built on PostgreSQL. A relational system is vital to handle the interconnected network of user profile variables, recursive pastoral care links, hierarchical department bounds, and strict sequence triggers for the course tracking engines.



4.1.1 Structural Entity-Relationship Diagram (ERD) – Text Description

The ERD consists of the following main entities and relationships:

- **users** (central entity) – contains profile, authentication, spiritual mentorship fields, and course submission references.
- **service_classes** – top-level departments (e.g., Education, Choir, Member Affairs).
- **sub_classes** – child units under a service class, with foreign key parent_class_id referencing service_classes(id).
- **lms_batches** – cohorts (ዙረ) for a specific course track (Gubae Abew, Hawaryat, Ecclesiae).

- **lms_enrollments** – junction table linking users and lms_batches, storing progress (including references to manually implemented lesson page URLs).
- **course_submissions** – table to track teacher content submissions, statuses (Draft, Submitted, Under Review, Approved, Rejected, Implementation in Progress, Published), feedback, and references to the final implemented web page URLs.

Relationships:

- users.service_class_id → service_classes(id) (many-to-one, nullable for USER).
- users.repentance_father_id, etc. → users(id) (self-referential, nullable).
- service_classes → sub_classes (one-to-many, cascade delete).
- sub_classes.sub_chair_id, etc. → users(id) (nullable).
- lms_enrollments.user_id → users(id), batch_id → lms_batches(id) (many-to-many).
- course_submissions.user_id → users(id) (teacher), course_submissions.batch_id → lms_batches(id).

4.1.2 Relational Cardinalities & Mapping Logic

- **Self-Referential Mentorship Ties (1:N):** A student row points to four distinct mentor profiles via foreign keys; deleting a mentor account triggers SET NULL.
- **Service Classes vs. Sub-Classes (1:N Strong Dependency):** One department owns zero to many sub-units; dropping a master service class cascades delete to sub-classes.
- **LMS Tracking Core (M:N via Junction):** lms_enrollments joins Users and Batches.
- **Course Submissions (1:N):** One teacher can have many submissions; each submission belongs to one batch and course track.

4.2 Data Dictionary

4.2.1 Table: users

Contains profiles, institutional properties, and security access identifiers.

Column Name	Data Type	Key Type	Nullable	Default / Constraints	Description
id	UUID	PK	NO	gen_random_uuid()	Immutable system-generated identity token.
full_name_three_parts	VARCHAR(150)	None	NO	None	Combined First, Father's, and Grandfather's names.

					Format: 'FirstName FatherName GrandfatherName' (space-separated).
email	VARCHAR(100)	None	NO	UNIQUE	Primary login identifier.
password_hash	VARCHAR(255)	None	NO	None	bcrypt hash (work factor 12).
university_id	VARCHAR(50)	None	YES	UNIQUE	Official campus student ID. Nullable until Tier 2.
academic_dept	VARCHAR(100)	None	YES	None	Registered branch of study. Nullable until Tier 2.
academic_year	INT	None	YES	CHECK (1–5)	Academic tier (1 to 5). Nullable until Tier 2.
dorm_block	VARCHAR(20)	None	YES	None	Assigned housing block. Nullable until Tier 2.
dorm_room	VARCHAR(20)	None	YES	None	Room number. Nullable until Tier 2.
phone_number	VARCHAR(20)	None	YES	None	Optional contact number for WhatsApp/urgent notifications.
profile_image_url	VARCHAR(255)	None	YES	'/assets/avatar.png'	URL to compressed image (<100 KB).
bio	TEXT	None	YES	None	Freeform personal description.
system_role	ENUM	None	NO	'USER'	Allowed: SECRETARIAT_CHAIRMAN, SECRETARIAT_VICE, SECRETARIAT_SECRET

					ARY, SERVICE_MANAGER, TEACHER, MEMBER, USER.
service_class_id	UUID	FK	YES	REFERENCES service_classes(id)	Department assignment. Null only for USER or pre-verified.
repentance_father_id	UUID	FK	YES	REFERENCES users(id)	Confession Priest.
repentance_deacon_id	UUID	FK	YES	REFERENCES users(id)	Coordinator Deacon.
spiritual_father_id	UUID	FK	YES	REFERENCES users(id)	Senior brother mentor.
spiritual_mother_id	UUID	FK	YES	REFERENCES users(id)	Senior sister mentor.
created_at	TIMESTAMP	None	NO	CURRENT_TIMESTAMP	Account creation time.

4.2.2 Table: service_classes

Column Name	Data Type	Key Type	Nullable	Default / Constraints	Description
id	UUID	PK	NO	gen_random_uuid()	Primary key.
class_name_amharic	VARCHAR(100)	None	NO	UNIQUE	e.g., 'ትምህርት ክፍል', 'Default Class'.
is_public_registration	BOOLEAN	None	NO	TRUE	If FALSE, blocks user selection.

4.2.3 Table: sub_classes

Column Name	Data Type	Key Type	Nullable	Default / Constraints	Description
id	UUID	PK	NO	gen_random_uuid()	Primary key.
parent_class_id	UUID	FK	NO	REFERENCES service_classes(id) ON DELETE CASCADE	Master

				N DELETE CASCADE	department
sub_class_name	VARCHAR(100)	None	NO	None	e.g., 'ዕለተ ሰንበት ንዑስ ክፍል'.
sub_chair_id	UUID	FK	YES	REFERENCES users(id) ON DELETE SET NULL	Sub-class leader.
sub_vice_id	UUID	FK	YES	REFERENCES users(id) ON DELETE SET NULL	Assistant leader.
sub_secretary_id	UUID	FK	YES	REFERENCES users(id) ON DELETE SET NULL	Recorder.

4.2.4 Table: lms_batches (ቱር)

Column Name	Data Type	Key Type	Nullable	Default / Constraints	Description
id	UUID	PK	NO	gen_random_uuid()	Primary key.
course_track	ENUM	None	NO	None	GUBAE_ABEW, GUBAE_HAWARYAT, GUBAE_ECCLESIAE.
batch_number	INT	None	NO	None	Incremental (e.g., 3 for ቱር 3).
status	ENUM	None	NO	'ACTIVE'	ACTIVE or GRADUATED. Only one ACTIVE per track.

4.2.5 Table: lms_enrollments

Column Name	Data Type	Key Type	Nullable	Default / Constraints	Description
id	UUID	PK	NO	gen_random_uuid()	Primary key.
user_id	UUID	FK	NO	REFERENCES users(id) ON DELETE CASCADE	Student.
batch_id	UUID	FK	NO	REFERENCES lms_batches(id) ON DELETE CASCADE	Batch.
completed_lessons	JSONB	None	NO	'[]'	Array of completed

					lesson IDs or manually implemented page URLs.
quiz_scores	JSONB	Non e	NO	'{}'	Key-value store of quiz scores.
final_exam_score	NUMERIC(5,2)	Non e	YES	None	Teacher-entered grade.
is_passed	BOOLEAN	Non e	NO	FALSE	Completion flag.
certificate_hash	VARCHAR(255)	Non e	YES	UNIQUE	Verification string for certificate.

4.2.6 Table: course_submissions (new for manual workflow)

Column Name	Data Type	Key Type	Nullable	Default / Constraints	Description
id	UUID	PK	NO	gen_random_uuid()	Primary key.
teacher_id	UUID	FK	NO	REFERENCES users(id) ON DELETE CASCADE	Teacher submitting content.
batch_id	UUID	FK	NO	REFERENCES lms_batches(id) ON DELETE CASCADE	Target batch.
title	VARCHAR(200)	Non e	NO	None	Lesson or module title.
content_package	TEXT	Non e	NO	None	Description or links to submitted materials (could be JSON or plain text).
status	ENUM	Non e	NO	'DRAFT'	DRAFT, SUBMITTED, UNDER_REVIEW, APPROVED, REJECTED,

					IMPLEMENTATION_IN_PROGRESS, PUBLISHED.
review_feedback	TEXT	None	YES	None	Feedback from Education Manager if rejected.
implemented_page_url	VARCHAR(500)	None	YES	None	URL of the manually created web page after technical implementation.
submitted_at	TIMESTAMP	None	YES	None	When teacher submitted.
reviewed_at	TIMESTAMP	None	YES	None	When Education Manager reviewed.
implemented_at	TIMESTAMP	None	YES	None	When Technical Team completed implementation.
published_at	TIMESTAMP	None	YES	None	When made visible to students.

4.3 Reports

4.3.1 Centralized Master Membership Document

- **Target Consumer:** Member Affairs (አገልግ ፖዲይ) Administrator and SECRETARIAT_CHAIRMAN.
- **Output Format:** Sortable UI data view frame supporting secure CSV extractions.
- **Aggregated Components:** Student Full Name, University ID, Engineering/Science Department, Year, Block/Room location data, assigned Service Department, and identity tracking parameters for their personal Confession Father, Coordinator Deacon, and Spiritual Mother and Father.
- **Security Guard:** Strict role-based validation. Access is limited strictly to Member Affairs Managers and the Secretariat Chairman; all other internal query variants are immediately intercepted and terminated with a permissions failure error.

4.3.2 Scoped Departmental Roster Summary

- **Target Consumer:** Departmental Service Managers (e.g., Education Manager, Mezmur Manager).
- **Output Format:** Clean functional roster table visible on mobile dashboards.

- **Aggregated Components:** Profile image avatar, full student name, internal sub-class team assignments (e.g., Youth Choir), local leadership status roles (Sub-Chair, Sub-Vice, Sub-Secretary), personal text bios, and active communication contacts, etc.
- **Security Guard:** Secure tenant boundaries. The application engine filters queries using the manager's active `service_class_id` context. Managers cannot run queries to pull data across departments.

4.3.3 Term Completion Ledger & Grade Grid

- **Target Consumer:** Evaluating Course Teachers and the Education Manager.
- **Output Format:** Tabular spreadsheet layout organized by active Batch ID (批号).
- **Aggregated Components:** Roster name records, calculated automated quiz scores, open-ended essay download routes, manual grading fields, feedback message files, and final `is_passed` parameter status changes.

4.4 Data Acquisition, Integrity, Retention, and Disposal

4.4.1 Image Optimization & Zero-Blob Infrastructure

To protect server transaction speeds and run the system without web storage hosting costs:

- **The Binary Block Rule:** Image files, PDF documents, and streamable videos cannot be uploaded as large binary objects directly into PostgreSQL table arrays.
- **Frontend Resizing Engine:** Profile photo submissions pass through a client-side JavaScript canvas engine that flattens file formats down to WebP, forces safe dimensions, clips file weight boundaries below **<100 KB**, and pushes them to cloud storage before mapping the text URL string back to the database user record.
- **Cloud Embedding Handshakes:** Educational library records, text sheets, and video segments are logged within schemas as text link references. The interface frames inject these links dynamically via YouTube or Google Drive viewing embeds.
- **Manual Course Implementation (No Visual Builder):** The system strictly rejects the usage of an automated Visual Course Builder. For course delivery, the Teacher submits the course content to the Education Class. Once the class verifies the curriculum, it is sent to the Technique Team, who manually implement it as a structured web page and include it in the designated course delivery sector. The database simply stores the target routing links to these customized web pages.

4.4.2 Database Relational Integrity Safeguards

- **Structural Cascades:** Deleting a `sub_classes` team entity triggers a SET NULL routine across the user leadership fields (`sub_chair_id`, `sub_vice_id`, `sub_secretary_id`). This cleanly dissolves internal operational nodes without corrupting or modifying primary student records in the users data grid.

- **LMS Progress Verification & Open Access:** Because the foundational *Gubae Abew* course is open to anyone signed up in Tier 1 (Public Access Account), the system requires absolutely no membership verification to access its specific course pages. However, backend API interceptors strictly check a student's tracking records before accepting new enrollments into advanced tracks. Registration requests for *Gubae Ecclesiae* or *Gubae Hawaryat* are blocked unless the ledger confirms `is_passed: true` for preceding foundational courses and the user has passed formal membership checks.

4.4.3 Data Retention, Archiving, and Account Disposal

- **Graduation State Freezing:** When a course batch moves to GRADUATED, the application backend disables all incoming write mutations against its children nodes. This locks student scoring sets into a permanent, unalterable ledger for future diocese auditing.
- **Alumni Profile Archiving:** Student user profiles are never hard-purged when they graduate from the university or conclude active service. Instead, they transition into an ARCHIVED_ALUMNI state, moving out of operational management views while retaining their historical records in the reporting log.
- **Administrative Hard Deletions:** If the Member Affairs Administrator or Secretariat Chairman determines an account entry is fraudulent or unauthorized, the interface provides a permanent hard-purge button. Running this command clears the user row and cascades across pending notifications, log markers, and comment collections, completely wiping out trace metadata.

4.5 Data Migration Requirements

FR-MIG-01 [P0]: The system must provide utilities to import:

- The pre-seeded graduate ledger (*Gubae Abew* completion list) from CSV/Excel provided by Member Affairs.
- Existing member profiles from legacy paper or digital records (format TBD with Member Affairs).

FR-MIG-02 [P0]: Migration scripts must validate email uniqueness, university ID format, and required fields before insertion.

FR-MIG-03 [P1]: All import operations must be logged with timestamp, user who performed import, count of records processed, and count of failures.

FR-MIG-04 [P0]: Database snapshots must be taken before any bulk migration to enable rollback.

5. Quality Attributes

5.1 Performance

NFR-PERF-01 [P0]: Public read operations must complete in < 200 ms under nominal load. Nominal load is defined as 500 concurrent authenticated sessions with average 20 requests/minute, running on baseline hardware of 2 vCPU, 4GB RAM.

NFR-PERF-02 [P0]: Complex write transactions (Tier 2 submission, JSONB updates) must complete in < 500 ms.

NFR-PERF-03 [P1]: Initial page load must be < 1.5 MB compressed; lazy loading implemented.

NFR-PERF-04 [P1]: Cumulative Layout Shift (CLS) must be < 0.1.

5.2 Security

NFR-SEC-01 [P0]: Passwords hashed with bcrypt (work factor 12) before storage.

NFR-SEC-02 [P0]: JWTs transmitted in HttpOnly, Secure, SameSite=Strict cookies.

NFR-SEC-03 [P0]: OWASP Top 10 compliance; parameterized ORM queries to prevent SQL injection.

NFR-SEC-04 [P0]: Global input validation pipelines strip unrecognised fields, sanitise HTML, enforce length constraints.

5.3 Usability

NFR-USAB-01 [P0]: 44×44px touch targets, two-click rule for primary actions.

NFR-USAB-02 [P1]: Amharic/English toggle; Noto Sans Ethiopic; line-height ≥1.6 for Ethiopic.

NFR-USAB-03 [P1]: Inline tooltips and placeholders for complex administrative tasks.

5.4 Accessibility (WCAG 2.1 AA)

NFR-ACC-01 [P0]: The system must conform to Web Content Accessibility Guidelines (WCAG) 2.1 Level AA.

NFR-ACC-02 [P0]: All functionality must be operable through a keyboard interface without requiring specific timing.

NFR-ACC-03 [P0]: All non-text content must have text alternatives. ARIA labels must be present on interactive elements.

NFR-ACC-04 [P0]: Text and interactive elements must meet minimum contrast ratio of 4.5:1 for normal text and 3:1 for large text.

NFR-ACC-05 [P0]: The interface must support browser text zoom up to 200% without loss of content or functionality.

5.5 Reliability & Availability

NFR-REL-01 [P0]: 99.5% uptime (excluding scheduled maintenance).

NFR-REL-02 [P1]: Client-side resilience: preserve cache and auto-retry failed requests with exponential backoff.

NFR-REL-03 [P0]: Daily encrypted snapshots stored in cold isolation.

NFR-REL-04 [P0]: RPO $\leq$ 24 hours, RTO $<$ 4 hours.

5.6 Maintainability & Scalability

NFR-MAIN-01 [P1]: Clean separation of concerns (entities, use cases, delivery frameworks).

NFR-MAIN-02 [P0]: Linear database scaling due to zero-blob architecture.

NFR-MAIN-03 [P1]: Immutable verification logs for audit.

5.7 Observability & Audit

NFR-LOG-01 [P1]: Centralized error monitoring with real-time alerting for unhandled exceptions.

NFR-LOG-02 [P0]: PII (passwords, emails, dorm numbers) must never appear in plaintext logs.

NFR-LOG-03 [P0]: Immutable audit records for permission changes, role upgrades, content state changes, manual overrides.

NFR-LOG-04 [P0]: Each record includes timestamp, user ID, source IP, and pre/post state delta.

5.8 Legal Compliance

FR-LEGAL-01 [P0]: The system must comply with applicable Ethiopian data protection laws and internationally recognized standards for student data.

FR-LEGAL-02 [P0]: University ID numbers and academic information must be treated as sensitive personally identifiable information (PII). Access logs must be maintained for any administrative view of this data.

FR-LEGAL-03 [P0]: The system must present a Terms of Service and Acceptable Use Policy during Tier 1 registration. User must explicitly accept before account creation.

FR-LEGAL-04 [P0]: The system must include a checkbox affirming the user is at least 18 years old or has parental consent, as required for university students.

6. External Interface Requirements

6.1 User Interfaces

6.1.1 Design System Choices

- **Design Tokens:** Custom Tailwind CSS theme with Orthodox Tewahedo liturgical colors (burgundy/crimson primary, gold/amber secondary).
- **Component Specifications:** UI components must follow headless component patterns with accessible ARIA labeling and 44x44px touch targets.

6.1.2 Layout & Responsiveness Expectations

- Breakpoints: mobile (320-480px) single-column, tablet (481-1024px) collapsible drawers, desktop (1025px+) fixed sidebars.
- Mobile admin screens use tab-routing instead of dense grids.

6.1.3 Role-Specific Dashboard Layout Interfaces

(UI layout only; see Section 3.7.2 for privileges.)

A. Unauthenticated / Public Visitor Interface

Unauthenticated visitors can browse public content (About page, Public Announcements, Public Library). Authentication (Tier 1) is required to access any member-only features or the Gubae Abew course pages. The root URL shows a landing page with registration/login options.

B. Secretariat Chairman Dashboard

Global governance panel, executive assignment portal, role transfer utility, read-only financial/member log block.

C. Secretariat Vice & Secretary Interface

Announcement composition studio, global library CRUD grid, read-only member census.

D. Universal Shared Features (All Departmental Roles)

- *Internal Hierarchy & Sub-Class Delegation Module*: Create sub-classes, assign sub-class leaders, allocate members, view roster.
- *Scoped Departmental Announcement Studio*: Targeted internal announcements.
- *Unified Plan & Report Matrix*: Upload and view annual plans (ዕቅዶች) and quarterly reports (ሪፖርቶች).

E. Teacher Dashboard

- **Course Submission Interface**: Form to submit course content (title, description, attached documents/video links, quiz definitions). Shows submission history and status (Draft, Submitted, Under Review, Approved, Rejected, Implementation in Progress, Published).
- **Grading Dashboard**: For manually grading open-ended exam submissions.

F. Member Fellowship Interface

- Scoped news feed (public + department).
- **LMS Phase Hub**: Shows Gubae Abew (always accessible), Gubae Hawaryat, and Gubae Ecclesiae (only if member and prerequisites met). For Gubae Abew, any USER can access; for Phases 2 & 3, access restricted to MEMBER role with completed prerequisites.
- Pastoral care index card.

6.2 Software Interfaces

6.2.1 Frontend-to-Backend REST API Handshake

- JSON payloads; multipart only for profile images.
- Standardised error response schema (status, message, error type, timestamp, path).
- API documentation must be generated from code annotations and accessible to administrators at a documented endpoint (protected from anonymous access).

6.2.2 YouTube IFrame Player API

FR-EXT-01 [P0]: System must embed YouTube videos via IFrame API with autoplay disabled and related videos restricted.

6.2.3 Google Drive API Document View Frame

FR-EXT-02 [P0]: PDF and document resources embedded using Google Drive preview iframe with "Anyone with link can view" permission.

6.3 Hardware Interfaces

NFR-HW-01 [P0]: No device-specific hardware integration; runs in standard web browsers with hardware-accelerated rendering.

6.4 Communications Interfaces

NFR-COMM-01 [P0]: TLS 1.2/1.3 required; HTTP redirected to HTTPS.

NFR-COMM-02 [P0]: CORS whitelist based on environment variables.

NFR-COMM-03 [P2]: Future WebSocket/SSE push notifications possible but not required for MVP.

7. Quality Attributes

Note: This section consolidates additional quality attributes not already covered in Section 5. However, to avoid duplication, the primary quality attributes are in Section 5. The following are kept for reference but stakeholders should refer to Section 5 for authoritative requirements.

7.1 Safety

NFR-SAFE-01 [P0]: Pastoral privacy: Service Managers sandboxed by department.

NFR-SAFE-02 [P1]: Graceful degradation on external API failures.

NFR-SAFE-03 [P0]: ACID transactions for multi-step mutations; daily encrypted backups retained 30 days.

7.2 Maintainability and Scalability

NFR-MAIN-01 [P1]: Clean separation of concerns (entities, use cases, delivery frameworks).

NFR-MAIN-02 [P0]: Linear database scaling due to zero-blob architecture.

NFR-MAIN-03 [P1]: Immutable verification logs for audit.

(Accessibility requirements are in Section 5.4; they are not duplicated here.)

8. Internationalization and Localization Requirements

8.1 Multilingual Core Framework

To serve the diverse linguistic needs of the university community and honor the liturgical tradition of the Church, the system must implement an internationalization (i18n) layer supporting runtime translation toggles across two target languages: **Amharic, and English..**

Language	Scope of Application	UI Handling Character Set
Amharic	Primary Administrative Default interface, Official Announcements, Core LMS Instruction.	UTF-8 Ethiopic/Ge'ez Unicode extensions.
English	Secondary UI Alternative, Academic Resource Titles, Technical Log Sheets.	UTF-8 Standard ASCII.

8.2 Font and Rendering Optimization

- **Typography Guards:** To completely eliminate empty text character boxes ("tofu blocks") on older mobile phone viewports, the application frontend must dynamically bundle and serve open-source web fonts supporting Ethiopic scripts, specifically **Noto Sans Ethiopic** for general UI text structures and **Abyssinica SIL** for classical Ge'ez rendering.
- **Line-Height Budgets:** The styling engine must programmatically increase standard line-height configurations by a factor of at least 1.6 whenever an Ethiopic character locale is selected. This adjustment is necessary because Ge'ez script variants possess intricate vertical serifs that blend awkwardly and cause eye strain on small screens if standard Latin text line dimensions are used.

8.3 Locality, Dates, and Calendar Standards

- **Dual Calendar Engine:** The system backend must process and log all temporal states (e.g., created_at parameters) using standard UTC ISO-8601 Gregorian timestamps. However, the client presentation layer must feature an algorithmic conversion utility to display localized date arrays utilizing the **Ethiopian Calendar (E.C.)** tracking format across liturgical schedules, official circulars, and course batches (ዓ.ም.).
- **Time System Presentation:** The application interface must allow users to toggle their display preferences between the standard Gregorian 24-hour notation format and the traditional East African 12-hour day/night cycle format (where hour zero corresponds to 6:00 AM Gregorian time).

9. Other Requirements

9.1 Ecclesiastical Regulatory Compliance

- **Theological Doctrine Safeguard:** The application is not a generalized open forum. In accordance with the internal guidelines (የውስጥ ደንብ) of the Enda Eyesus Gbi Gubae and the administrative oversight of the Diocese (ሀገረ ስብከት), no public theological teaching material, dogma lesson block, or liturgical analysis may be pushed to a Published status without an explicit approval signature from the **Education Department Manager** logged in the system.
- **Hierarchical Oversight Portals:** The system audit log must provide an immutable read-only view route exportable via secure tokens. This mechanism allows the executive committee (ሥራ አስፈጻሚ ፖለቲካ) or visiting representatives from the Diocese coordinator network to verify system audit metrics without editing active production data grids.

9.2 Campus Network Resiliency Constraints (Intermittent Offline Tolerance)

NFR-OFFLINE-01 [P1]: The system must support offline read access for LMS text modules and library metadata using client-side caching.

NFR-OFFLINE-02 [P1]: On network loss, serve cached content and queue non-critical updates until connectivity returns.

10. Training and Documentation Requirements

FR-DOC-01 [P1]: The system must include context-sensitive help (tooltips and inline guides) for all administrative functions.

FR-DOC-02 [P1]: A separate administrator guide must be provided covering: user management, batch lifecycle, announcement publishing, report generation, and the manual course submission/implementation workflow.

FR-DOC-03 [P2]: Video tutorials (5-10 minutes each) must be created for: Student Registration, Member Affairs workflow, Teacher Course Submission process, Batch graduation for Education Manager.

FR-DOC-04 [P1]: Documentation must be available as downloadable PDF and within the application as a help section.

Appendices

Appendix A: Glossary

Institutional and Liturgical Nomenclature

- **Gbi Gubae (ግቢ ጉባኤ):** The University Orthodox Christian Student Fellowship movement organized under the canonical authority of the Ethiopian Orthodox Tewahedo Church Mahere Qidusan framework.
- **Gubae Abew (ጉባኤ አበው):** Phase 1 foundational training module within the fellowship, focusing on Church history, patristics, and traditional layout regulations. Completion of this track is an absolute prerequisite to becoming a verified full member.
- **Gubae Hawaryat (ጉባኤ ሐዋርያት):** Phase 2 intermediate academic path focusing on dogmatic theology, apologetics, and apostolic service strategies.
- **Gubae Ecclesiae (ጉባኤ ኤክሌሲያ):** Phase 3 advanced curriculum covering liturgical history, advanced ecclesiology, and leadership administration.
- **ZUR / Batch (ዙር):** A distinct chronological term instance tracking a specific cohort of university students moving through a course lifecycle within a specific academic year cycle.
- **Tsifet Bet / Secretariat (ጽሕፈት ቤት):** The core executive office of the fellowship, comprising the Chairman, Vice-Chairman, and General Secretary.
- **Yenisha Abat / Repentance Father (የንሕሐ አባት):** A canonical confessor priest assigned to provide sacramental guidance, oversight, and spiritual absolution to an individual Christian member.
- **Menfeseawi Abat / Enat (መንፈሳዊ አባት/እናት):** A designated senior spiritual mentor (brother or sister) within the fellowship assigned to support an underclassman student's moral and spiritual development.

Technical Framework Nomenclature

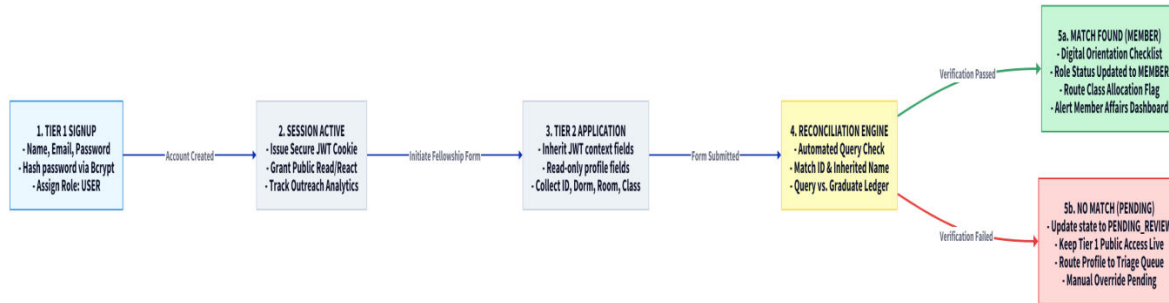
- **Zero-Blob Database Architecture:** A software engineering constraint dictating that no heavy binary resource layout files (such as raw images, PDFs, or MP4 files) are stored within the relational tables. All storage queries rely on indexed URL strings mapping to external asset targets.
- **JWT (JSON Web Token):** A compact, URL-safe means of representing claims to be transferred between two parties. Tokens are encrypted within server boundaries and passed to client view slots inside secure browser tracking layers.
- **RBAC (Role-Based Access Control):** A system permission validation methodology that checks a user's role parameters against access control records before resolving a requested transaction route.
- **JSONB:** A decomposed binary data storage format native to PostgreSQL that allows developers to write, index, and query unstructured JSON array formats directly within a stable relational environment.

Appendix B: Analysis Models

B.1 Member Registration and Triage Sequence

This layout maps the validation and routing architecture executed immediately after a new student account registration submission hits the system framework.

Technical Sequence Diagram



1. User completes Tier 1 sign-up → USER record created.
2. User initiates Tier 2 membership → provides university ID, department, year, dorm, preferred class.
3. System validates against pre-seeded Gubae Abew graduate ledger.
 - Match found:
 - a. Show orientation checklist.
 - b. User acknowledges.
 - c. System upgrades role to MEMBER.
 - d. Assigns service_class_id (preferred or Default Class).
 - e. Notifies Member Affairs Manager.
 - No match found:
 - a. Set status to PENDING_REVIEW.
 - b. Keep USER privileges.
 - c. Push profile to Member Affairs review queue.
 - d. Member Affairs manually verifies (approve/reject).
 - e. If approved → same as match found; if rejected → user remains USER with explanation.

Appendix C: Requirement Traceability Matrix

Requirement ID	Description	User Story	Test Case Reference	Status
FR-REG-01	Initial registration form	US-REG-01	TC-REG-01	Approved
FR-LMS-06	Manual course submission	US-LMS-06	TC-LMS-06	Approved

	workflow			
FR-LMS-09	Gubae Abew open access for USER	US-LMS-09	TC-LMS-09	Approved